



Emotet Malware

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Introduction

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Year: Pre-Junior

Major: Computer Science -> CST
(Security Concentration)

Fun fact: I play a lot of volleyball with friends in my free time.



Overview of the Hack

Emotet is a sophisticated malicious software that is used as a distributing source of the malware. It's a global malware that is said to have originated in Ukraine and was first detected in 2014 but has had many major events since then. One in January and November 2021, one in March 2023, and one early 2025.

This isn't a unique attack, but more so a type of Trojan that a cybercrime developed and pushed into the internet, reaching global networks. Emotet was crated for financial gain, as it was first designed as a banking Trojan. It evolved into a malware-as-a-service platform, which means other cybergroups used it to distribute more malware.

How do Emotet attacks happen?

Emotet attacks happen primarily through spam emails. They are designed to look legitimate but actually contain malicious attachment in a form of PDF, Word, or Excel documents and even links that direct you to a compromised website. Emotet continues to gain access to your device through registry keys, hidden files, and scheduled tasks. It can even potentially disable antivirus software and tools you may have and allow for more malwares to be introduced into your system.

There was also an abuse of MS App Installer protocol that were used in campaigns from November 2021 to January 2022.



Technical Details & IOC's

Looking through VirusTotal, I found an example of Emotet Malware (<https://www.virustotal.com/gui/file/3755718db9d33f4aba2563de454d4530a308b41b1096c904102d08e2101f2020/community>)

Local IP Address: 192.168.1.221

File Name/Malware Dropper: TsarBomba.exe

Type: Executable

C2 Server URL:
[hxxps://steamcommunity.com/profiles/76561199851454339](https://steamcommunity.com/profiles/76561199851454339)

C2 Server IP Address: 154.91.34.165:64951



JaffaCakes118

📅 21 days ago

File Info:

Filename:

TsarBomba.exe

Threat Score:

10/10

Family:

- chaos
- cobaltstrike
- dcrat
- discordrat
- gh0strat
- lockbit
- modiloader
- redline
- rokrat
- sectopratt
- umbral
- vidar
- wshrat

Botnet:

- cheat
- 158fdd2a4f5abb978509580715e5353f
- 987654321

C2:

- 154.91.34.165:64951
- hxxps://t.me/m00f3r
- hxxps://steamcommunity.com/profiles/76561199851454339
- hxxp://103.171.35.26:9443/dot.gif
- 192.168.1.221

File Report:

<https://jaffacakes118.dev/analysis/3755718db9d33f4aba2563de454d4530a308b41b1096c904102d08e2101f2020>
<https://tria.ge/250518-3g9myafj71>

Tags:

#chaos
#cobaltstrike
#dcrat
#discordrat
#gh0strat
#lockbit
#modiloader
#redline
#rokrat
#sectopratt
#umbral
#vidar
#wshrat
#158fdd2a4f5abb978509580715e5353f
#987654321
#cheat

MITRE ATT&CK Techniques

T1134 - Access Token
Manipulation: Token
Impersonation/Theft

T1547 - Boot or Logon
Autostart Execution

T1071-Application
Layer Protocol: Web
Protocols

T1543-Create or
Modify System
Process: Windows
Service

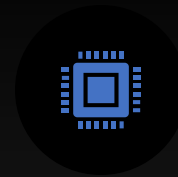
Affected Products & Services



NetPass.exe - Developed by NirSoft to recover passwords on the network that is stored for the current user logged in



Outlook scrapper - Takes names and email addresses from Outlook accounts



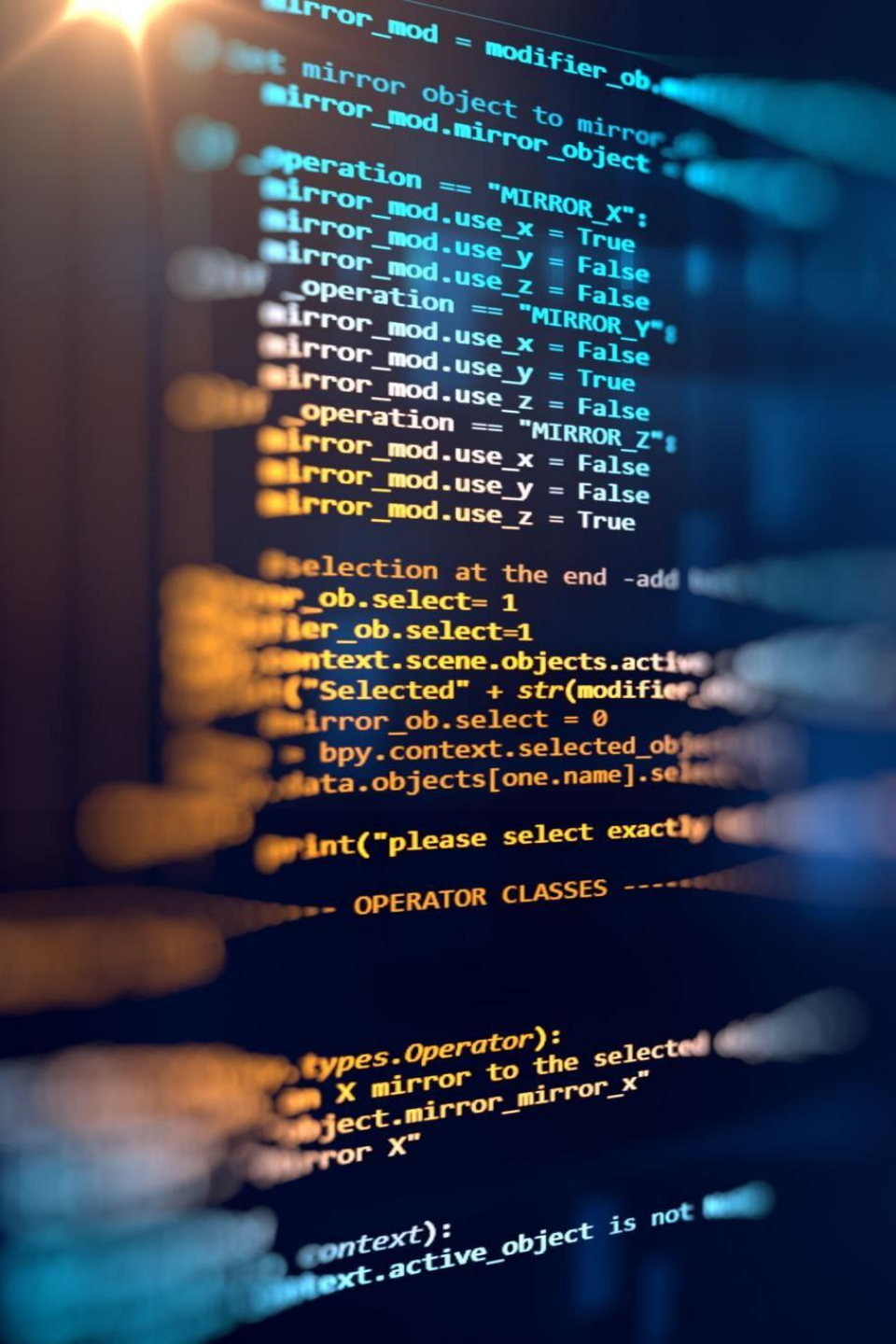
WebBrowserPassView - Password recovery tool that takes passwords that's stored by search engines like FireFox and Chrome and gives them to credential enumerator module



Credential enumerator - RAR file that self-extracts and contains a bypass and service component



PayPal - A recent campaign that imitates PayPal receipts and other forms of emails to make the user open the PDF, link, etc. on the email

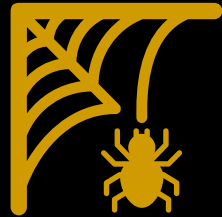


Mitigations

There are many methods of mitigating Emotet attacks

- Patching software: Always ensure you are running the latest version of all software you are using
- Remedying known vulnerabilities
- Only use secure networks and make sure to install on a private VPN network
- Developing an incident response plan that allows for you to deal with an attack and ensures as little damage as possible
- Requiring administrator credentials to install software: This ensures employees have to have the permissions to download any sort of applications

Hacker Identification



AS I MENTIONED BEFORE, THIS MALWARE
POINTS TO A CYBERGROUP CALLED MUMMY
SPIDER (A.K.A MEALYBUG, TA542, GOLD CABIN)
THAT ORIGINATED IN UKRAINE



THERE WERE TWO UNNAMED INDIVIDUALS
THAT WERE APART OF MUMMY SPIDER
ARRESTED IN UKRAINE BACK IN JANUARY
2021.

Repercussions of Hack

There is a big financial loss for companies and individuals that are infected by Emotet. There is a rough estimate of \$1 million dollars lost per incident and with it spreading globally since 2014, it's thought that Emotet has created billions of damages combined.

Certain infrastructure such as healthcare, government, private sectors, and educational institutions were targeted and were compromised.

Key Takeaways

I've noticed that with how big the internet has become, taking advantage of users using malware is much easier. There are so many methods of gaining access to data and devices that a simple click of your mouse button could send a virus coursing through your computer and the exposure of all your information. People are constantly finding ways to hack into systems for their own personal benefit and the internet is assisting them by giving them all the necessary knowledge they need. This is why new malwares are usually from unknown cybergroups that are formed.

Works Cited

<https://nvd.nist.gov/vuln/detail/CVE-2021-43890>

<https://www.cisa.gov/news-events/cybersecurity-advisories/aa20-280a>

<https://krebsonsecurity.com/2021/01/international-action-targets-emotet-crimeware/>

<https://www.hhs.gov/sites/default/files/emotet-the-enduring-and-persistent-threat-to-the-hph-tlpclear.pdf>

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<https://thehackernews.com/2024/06/authorities-ramp-up-efforts-to-capture.html>